Toronto MetRobotics
Science & Controls Team Application
Encoder Justification Report
Rahima N

After comparing all 8 encoders provided, I narrowed my choice down to either the AMT22 or the AMT11. As outlined, the robotic arm's design should prioritize accurate and repeatable motion. Upon comparing these encoders, these two seem to perform this task the best. Both encoders feature low power consumption and high resolution compared to the others. The AMT22 features 12- or 14-bit resolution, with an accuracy of 0.2°. Its high resolution allows the encoder to identify small changes in its angular position, and its accuracy reflects how well the measured position compares to the actual position. In contrast, the AMT11 offers selectable resolution, with its highest being as effective as the AMT22, with a similar accuracy as well.

Between these two encoders, I determined the most suitable option to be the AMT22. The primary difference between these two encoders is that the AMT22 is an absolute encoder, while the AMT11 is an incremental encoder. The AMT22 directly reports its angular position, allowing for easy position measurement from startup without calculating from a reference point like an incremental encoder. This seems more beneficial given the goal of this robotic arm where, in executing repeatable motion, the arm's angular position is an important attribute. Having easy access to this information simplifies tracking the joints' movements and any further adjustments needed to be made to execute instructions

Additionally, given the multiple joints of the arm, either a single-turn or a multi-turn configuration of the AMT22 can be used depending on the respective demand on each joint's range of motion. For simple joints that do not require multiple revolutions, a single-turn encoder can be applied, whereas joints that may need to undergo multiple revolutions can use a multi-turn AMT22. The AMT22 provides a more convenient alternative to the AMT11, despite being generally more expensive.